

LP3 (ISR) – Assignment 1 – Conflation Algorithm (RollNo: 43221)

Input Code:

```
import java.io.*;
import java.util.*;
public class Conflation {
    public static ArrayList<String> stopList = new ArrayList<String>();
    public static ArrayList<String> removestopList = new
ArrayList<String>();
    public static String suffixes[] = { "able", "ing", "ion", "y", "ment" };
    public static String stopwords[] = { "i", "big", "am", "m", "a", "we",
        "are", "it", "of", "this", "and", "is", "to", "at", "in", "was",
        "with", "doing", "It", "not", "our" };

    public static void conflation(String fname) {
        BufferedReader buff;
        int i = 0, j = 0;
        try {
            buff = new BufferedReader(new FileReader(fname));
            int flag = 0;
            String line = "";
            line = buff.readLine();
            String[] buffer = line.split(" ");
            for (i = 0; i < buffer.length; i++) {
                flag = 0;
                if (buffer[i].endsWith("."))
                    buffer[i] = buffer[i].replace(".", "");

                for (j = 0; j < stopwords.length; j++) {
                    if (buffer[i].equals(stopwords[j])) {
                        stopList.add(buffer[i]);
                        flag = 1;
                        break;
                    }
                }
                if (flag != 1 && !buffer[i].equals(null)) {
                    removestopList.add(buffer[i]);
                }
            }
            System.out.println("\n-----After Removing Stop Words-----");
            for (int k = 0; k < removestopList.size(); k++) {
                System.out.println(removestopList.get(k));
            }

            suffixesString(removestopList);
            countFrequency(removestopList);

        } catch (FileNotFoundException e) {
            e.printStackTrace();
        } catch (IOException e) {
            e.printStackTrace();
        }
    }

    private static void countFrequency(ArrayList<String> removestopList2) {
        // Mapping of String->Integer (word -> frequency)
        System.out.println("\n\n-----After Counting Frequency-----");
    }
}
```

```

        final Map<String, Integer> frequencyMap = new HashMap<String,
Integer>();
        for (int k = 0; k < removestopList.size(); k++) {
            String currentWord = removestopList.get(k);
            Integer frequency = frequencyMap.get(currentWord);
            // Add the word if it doesn't already exist, otherwise increment
the
            // frequency counter.
            if (frequency == null) {
                frequency = 0;
            }
            frequencyMap.put(currentWord, frequency + 1);
        }
        Iterator entries = frequencyMap.entrySet().iterator();
        while (entries.hasNext()) {
            Map.Entry entry = (Map.Entry) entries.next();
            String key = (String) entry.getKey();
            Integer value = (Integer) entry.getValue();
            System.out.println(key + " = " + value);
        }
    }
    private static void suffixesString(ArrayList<String> removestopList) {
        System.out.println("\n\n-----After Removing Suffixes-----
");
        for (int k = 0; k < removestopList.size(); k++) {
            String suffixString = removestopList.get(k);
            int flag = 0;
            for (int m = 0; m < suffixes.length; m++) {
                if (suffixString.endsWith(suffixes[m])) {
                    int len = suffixString.length();
                    int len1 = suffixes[m].length();
                    int len2 = len - len1;
                    String sufString = suffixString.substring(0, len2);
                    System.out.print(suffixString + "\t\t");
                    System.out.println(sufString);
                    flag = 1;
                    break;
                }
            }
            if (flag != 1)
                System.out.println(suffixString + "\t\t" + suffixString);
        }
    }
    public static void main(String[] args) {
        InputStreamReader st = new InputStreamReader(System.in);
        BufferedReader buff = new BufferedReader(st);
        String fname = "";
        System.out.println("Enter a filename:");
        try {
            fname = buff.readLine();
        } catch (IOException e) {
            e.printStackTrace();
        }
        conflation(fname);
    }
}

```

Output:

```
"C:\Program Files\Java\jdk-22\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition
2024.1.4\lib\idea_rt.jar=57545:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2024.1.4\bin" -
Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8 -Dsun.stderr.encoding=UTF-8 -classpath
F:\BE\LP3\GroupA_Assignment_1_Conflation\out\production\GroupA_Assignment_1_Conflation Conflation
```

Enter a filename:

sample.txt

-----After Removing Stop Words-----

This

sample

text

contains

various

words

sentences

The

purpose

text

demonstrate

the

functionality

the

conflation

algorithm

The

algorithm

will

remove

stop

words,

remove

suffixes,

count

the

frequency

each

word
will
provide
insights
into
the
text
analysis
process
Feel
free
modify
the
text
observe
the
results

-----After Removing Suffixes-----

This	This	
sample	sample	
text	text	
contains	contains	
various	various	
words	words	
sentences		sentences
The	The	
purpose	purpose	
text	text	
demonstrate		demonstrate
the	the	
functionality		functionalit
the	the	
conflation		conflat

algorithm	algorithm	
The	The	
algorithm	algorithm	
will	will	
remove	remove	
stop	stop	
words,	words,	
remove	remove	
suffixes,	suffixes,	
count	count	
the	the	
frequency		frequenc
each	each	
word	word	
will	will	
provide	provide	
insights	insights	
into	into	
the	the	
text	text	
analysis	analysis	
process	process	
Feel	Feel	
free	free	
modify	modif	
the	the	
text	text	
observe	observe	
the	the	
results	results	

-----After Counting Frequency-----

words, = 1

purpose = 1

conflation = 1
sentences = 1
functionality = 1
remove = 2
observe = 1
frequency = 1
The = 2
into = 1
This = 1
text = 4
free = 1
results = 1
algorithm = 2
process = 1
will = 2
insights = 1
Feel = 1
words = 1
count = 1
analysis = 1
sample = 1
each = 1
the = 6
modify = 1
contains = 1
stop = 1
provide = 1
various = 1
demonstrate = 1
suffixes, = 1
word = 1

Process finished with exit code 0